Acknowledgements

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Contents

Chapter 1

Introduction

1.1 Motivation

Financial time series data is never fully observed Limit-order book feeds drop quotes during bursts of order cancellations. Equity markets suspend trading when prices move too violently. Over-the-counter bond markets leave entire yield-curve tenors unquoted for days. Implied volatility surfaces contain large regions where no liquid option trade has occurred. Macro and alternative data arrive at mixed, irregular frequencies, with publication lags that create structural gaps at the daily level. This missing data is pervasive and often elusive, and gets in the way of advanced analysis. Thus practitioners or a researchers face the same decision: what value should be attributed to the missing observation, and how much uncertainty should be attached to that attribution?

The simplest and most common solution is to carry forward the last observed value, but this can lead to systematic mispricing, underestimation of risk, and distortion of correlations among assets.

We aim to take a different path, one that looks for solutions that aim to preserve as much as possible the statistical properties of the series, in order to maintain the integrity of the underlying data and consequently the robustness of any downstream model that relies on it. Since this can be considered an inference problem, we will also explore ML algorithms, and in classical machine learning approach, we will focus on quality of inference rather than interpretability of the model.

In this thesis we will test these different methods of imputation, and evaluate them not only on the accuracy of the imputation itself (MSE,...), but also on the quality of the downstream model that uses the imputed data, and how far it deviates from the case where all data is observed.

1.2 Literature review

The literature on missing data is vast and spread across statistics, econometrics and, more recently, machine learning. Rather than attempting an exhaustive survey, we trace the ideas this thesis directly builds on: the foundational framework that defines what missingness is, the classical likelihood-based methods with the EM algorithm at their core, the progressive shift towards algorithmic and deep learning imputers, and finally how these tools have been absorbed by economics and finance.

1.2.1 Foundations

Until the mid-1970s missing data was mostly treated as a nuisance to be engineered away: incomplete records were dropped, or filled in with some convenient value, and the analysis proceeded as if nothing had happened. The turning point is Rubin (1976), who proposed to treat missingness itself as a random object. Given a data matrix and an indicator matrix flagging which entries are observed, the joint distribution factorizes into the law of the data and the law of the *missingness mechanism*, the conditional distribution of the indicators given the data. Rubin's classification of this mechanism is still the standard language of the field: data are missing completely at random (MCAR) when the mechanism is independent of the data; missing at random (MAR) when it depends only on the observed part; and missing not at random (MNAR) otherwise. The central result of the paper is *ignorability*: under MAR, plus a parameter-distinctness condition, likelihood-based and Bayesian inference can ignore the mechanism altogether and work with the observed-data likelihood. This single result justifies most of the methods in this thesis; whenever it fails, as in the stressed missingness pattern of Section 2.1.1, every method that relies on it stands on shaky ground.

The monograph of Little and Rubin (2019), first published in 1987, organized the field around this framework and remains its canonical reference. Their taxonomy of approaches (complete-case analysis, weighting, single imputation, likelihood-based methods) is still the way the subject is taught, and their analysis of the simple methods is directly relevant here: listwise deletion is unbiased only under MCAR and can discard a dramatic fraction of the sample, while unconditional mean imputation distorts variances and correlations even under MCAR. These are precisely the two baselines we adopt in Chapter 2, and the literature predicts their failure modes quite exactly.

Two other foundational strands deserve mention. The first is weighting: instead of imputing, one reweights the complete cases by the inverse of their probability of being observed. Robins et al. (1994) developed this idea into a full semiparametric theory, including doubly robust estimators; Seaman and White (2013) give an accessible review. The second is *multiple imputation*, introduced by Rubin (1987): draw several imputed datasets from the posterior predictive distribution of the missing values, analyse each one as if it were complete, and combine the results with simple rules that propagate the imputation uncertainty into the final standard errors (see also Rubin, 1996). Multiple imputation is the historical answer to a criticism that applies to every single-imputation method in this thesis: an imputed dataset treated as real understates uncertainty, because imputed values are estimates, not observations.

The MNAR case is fundamentally harder, since the mechanism is not identifiable from the observed data alone and any analysis rests on untestable assumptions. In econometrics the problem was attacked early and largely independently by Heckman (1979), whose sample-selection model corrects for non-randomly missing outcomes through an explicit model of the selection process; it remains the standard MNAR tool in applied economics. In biostatistics the accepted practice is sensitivity analysis across plausible mechanisms (Molenberghs and Kenward, 2007). Finally, the translation of MAR to dependent, dynamically evolving data is more subtle than it appears; Commenges and Jacqmin-Gadda (2022) give a rigorous stochastic-process formulation, which is the natural setting for financial time series.

1.2.2 Likelihood methods and the EM algorithm

Once ignorability is granted, inference reduces to maximizing the observed-data likelihood, which almost never has a closed form because the missing entries must be integrated out. The Expectation–Maximization algorithm of Dempster et al. (1977) is the fundamental tool for this task, and arguably the most important paper in the field after Rubin (1976). Its logic mirrors the structure of the problem: the E-step computes the conditional expectation of the complete-data log-likelihood given the observed data and the current parameters, and the M-step maximizes it. The paper proved that each iteration increases the observed likelihood, unified a large number of pre-existing special-purpose schemes, and named the method. The convergence theory was completed by Wu (1983), who fixed a flaw in the original proof and gave the regularity conditions under which EM converges to a stationary point. Since EM never forms the observed information matrix, standard errors require extra work; Louis (1982) showed how to recover them from complete-data quantities. Among the many refinements that followed, the ECM and ECME algorithms (Liu and Rubin, 1994; Meng and Rubin, 1993) replace the M-step with cheaper conditional maximizations; McLachlan and Krishnan (2008) is the standard book-length treatment.

For our purposes the relevant instance is EM for the mean vector and covariance matrix of an incomplete multivariate normal sample (Little and Rubin, 2019, Ch. 11): the E-step reduces to linear regressions of the missing coordinates on the observed ones, and the fixed point delivers both the maximum likelihood estimate of $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and, as a by-product, conditional-expectation imputations. Since daily returns are decidedly non-Gaussian (Cont, 2001), the robust variant matters: Little (1988) derived EM for the multivariate Student-t, whose E-step simply downweights outlying observations, and Liu and Rubin (1995) made its estimation practical via ECME. This is the version we implement in Chapter 2.

The time series counterpart of this machinery is the state-space framework. The Kalman filter (Kalman, 1960) accommodates missing observations with no modification at all: when a measurement is absent, the update step is skipped and the prediction carries through. Jones (1980) exploited this to fit ARMA models to incomplete series by exact maximum likelihood, and Shumway and Stoffer (1982) closed the loop with EM: the Kalman smoother computes exactly the conditional expectations the E-step requires, so parameter estimation and imputation come out of the same iteration. Harvey and Pierse (1984) applied these ideas to economic time series, Kohn and Ansley (1986) settled exact interpolation for ARIMA models, and the monographs of Harvey (1989) and Durbin and Koopman (2012) remain the references for the general theory.

1.2.3 From multiple imputation to machine learning

On the applied side, the 1990s and 2000s standardized imputation practice. Schafer (1997) provided workable algorithms for multiple imputation under joint multivariate models, and the chained equations approach of van Buuren and Groothuis-Oudshoorn (2011) replaced the joint model with a sequence of univariate conditional models, one per variable, cycled until convergence; its flexibility made it the de facto standard in biostatistics and the social sciences. van Buuren (2018) gives a modern account of the whole practice.

In parallel, a more algorithmic tradition developed methods that dispense with an explicit probability model. k-nearest-neighbours imputation was popularized by Troyanskaya et al. (2001) for gene expression data and transfers directly to return panels: a missing entry is filled with a weighted average of the corresponding entries of similar rows (dates, in our case); random forests followed (Stekhoven and Bühlmann, 2012). A distinct and elegant line is low-rank matrix completion (Candès and Recht, 2009; Mazumder et al., 2010), which recovers missing entries under the assumption that the data matrix is approximately low-rank, a natural assumption for equity returns where a few factors explain most of the covariance.

Deep learning entered the field once recurrent networks were adapted to irregular inputs. GRU-D (Zhengping Che and Liu, 2016) feeds the missingness mask and the time since the last observation into a gated recurrent unit with learned decay; BRITS (Cao et al., 2018) treats imputations as variables of a bidirectional recurrent graph, trained by consistency between the two directions, and requires no assumption on the generating process. Generative approaches recast imputation as conditional sampling: GAIN adapts adversarial training (Yoon et al., 2018), GP-VAE combines a variational autoencoder with a Gaussian process prior in latent space (Fortuin et al., 2020), SAITS relies on self-attention (Du et al., 2023), and CSDI (Tashiro et al., 2021) uses conditional score-based diffusion, producing a full predictive distribution for the missing values rather than a point estimate, which is why we select it as our generative representative. Dong et al. (2024) survey this rapidly growing area, and Kreiss et al. (2024) bridge it with the signal processing literature.

Two caveats recur across this literature. First, essentially all deep imputers assume MAR, usually implicitly through their masked training objective; identification under more general mechanisms has been studied only very recently (Ma et al., 2025). Second, evaluation is almost universally pointwise, RMSE or MAE on artificially masked entries, which says little about whether the joint distribution, and in particular the covariance structure, survives imputation. Both caveats are starting points for this thesis.

1.2.4 Applications in economics and finance

In economics, missing data rarely goes by that name. The oldest example is non-synchronous trading: Scholes and Williams (1977) showed that stale prices, which are in effect missing prices carried forward, bias beta estimates, and derived corrected estimators. This is exactly the last-observation-carried-forward distortion described in Section 1.1.

The macroeconometric literature turned missing data handling into routine practice, because macro datasets are incomplete by construction: series are released at mixed frequencies and with staggered publication lags (the so-called ragged edge). Stock and Watson (2002) used an EM–principal-components iteration to extract factors from large unbalanced panels; Mariano and Murasawa (2003) treated quarterly GDP as a monthly series with two out of three values missing, estimated in state-space form; Giannone et al. (2008) formalized nowcasting, where the Kalman filter absorbs data releases as they arrive; and Doz et al. (2012) and Bańbura and Modugno (2014) developed maximum likelihood estimation of large factor models under arbitrary missingness patterns, essentially scaling Shumway and Stoffer (1982) to hundreds of series. This machinery runs in production at central banks today. For panels with a time dimension, Honaker and King (2010) adapted multiple imputation to time-series cross-section data, noting that off-the-shelf procedures ignore temporal smoothness.

In finance proper, the seminal reference is Stambaugh (1997), who considered assets whose return histories start at different dates, the typical shape of any real multi-asset panel, and derived maximum likelihood estimators of means and covariances that use the long histories to sharpen the moments of the short ones. The message, that discarding incomplete history throws away information about the covariance matrix, is a direct ancestor of our evaluation criteria. Kofman and Sharpe (2003) later imported multiple imputation into empirical finance, showing on several canonical applications that complete-case analysis materially biases the conclusions.

The topic has recently returned to the centre of asset pricing, driven by the firm-characteristics panels used in cross-sectional factor models, where the majority of entries can be missing. Bryzgalova et al. (2025) document that the profession's ad hoc fixes, typically cross-sectional medians, are not innocuous: the choice of imputation changes the estimated risk premia, and they propose a factor-based imputation exploiting both cross-sectional and time series dependence. In parallel, Freyberger et al. (2025) show that firm characteristics are missing *not* at random and develop estimators that remain consistent under such mechanisms. Both papers make, in the cross-section of characteristics, the point this thesis makes for return panels: what matters is not the imputation error itself but its effect on the downstream object of interest. Finally, Choudhury et al. (2021) highlight a specifically financial pitfall: imputing with a model trained on the full sample leaks future information into the imputed values, and they formalize the resulting trade-off between look-ahead bias and imputation variance.

Summing up, the statistical literature provides the taxonomy of mechanisms and the classical estimators, the machine learning literature provides increasingly powerful conditional models trained under an implicit MAR assumption, and the finance literature provides both the warning that missingness is often MNAR precisely when it matters most (crises, illiquidity) and the reminder that imputation should be judged by the downstream model. The experimental design of this thesis, controlled injection of MCAR and MNAR patterns into a complete equity panel, classical and deep imputers side by side, and evaluation on covariance geometry, risk measures and portfolio construction rather than pointwise error alone, sits deliberately at the intersection of these three strands.

1.3 Research questions

1.4 Contributions

1.5 Outline of the thesis

Chapter 2

Methodology

As already mentioned, missing data can arise in countless different ways, and the choice of methodology depends on the nature of the missingness, the type of data, and the specific goals of the analysis. We will focus on a subset of cases, namely those that involve imputation of the missing values with "information" from the whole dataset (no causal constraint), with in mind the goal of training a downstream model that is as close as possible to the one that would have been trained on the complete dataset. Our choice of data, models, and evaluation methods will be guided by these constraints. We will nevertheless mention when/how models should be tweaked to solve adjacent problems.

2.1 Data

As our main baseline, we chose the most studied and reliable dataset in finance: the end-of-day prices of the S&P500, in this case fetched with yfinance from 1/1/2010 to 30/5/2026 (potrei cambiare le date). It clearly is not a dataset inflicted with the curse of missing data; the only structural ones being occasional circuit breaks and listings/delistings from the index. To obtain a full dataset with no "real" missing data, we removed each stock that contained at least one missing observation. Furthermore, we selected a sample of 50 stocks from the index, stratified across industries; this is a sensible choice because:

1. with more 400 stocks the covariance matrix $\mathbf{\Sigma}$ quickly becomes unreliable (noisy covariances, skewed eigenvalues, unstable inversion. . .) as the number of observations shrinks;
2. it is easier to check whether the ground data is actually correct;
3. model training requires considerably less memory, in a time where RAM and GPU prices are skyrocketing;
4. it is well known that a well constructed portfolio with more than 30 stocks is already sufficiently diversified (since volatility scales with the inverse of the square root of

the number of securities).

This dataset will serve as the ground truth against which we will compare our results. In parallel, we will modify it by inserting fake missing observations according to different rules, as we define in the subsequent sections.

Now an important choice is whether to operate in the prices "space" P_t or the log returns space R_t. There are positives and negatives about each. For one, the raw dataset is prices, and in practice missing data will mostly come in the form of missing prices. Furthermore, in most applications we care slightly more about prices than returns, since the formers are what is actually bought and sold. But P_t has the drawback of not being a complete space ($p_{t,k} > 0$) and being highly level dependant, making the series non-stationary and very difficult for ML algorithms to handle. We will thus focus on R_t.

2.1.1 Missingness Injection

(Da scrivere meglio, questa è solo l'outline)

As outlined in the literature review ...

We used three algorithms to simulate different missingness patterns:

1. **Sparse**: each scalar $r_{t,k}$ is missing with probability r, independent of the rest of the dataset. This is the easiest example of MCAR and can simulate a randomly corrupted dataset;

2. **Contiguous gaps**: n points $r_{t,k}$ are chosen at random and eliminated. Then subsequent observations of the same variable $r_{t+1,k}$, ..., $r_{t+h,k}$ get deleted according to an exponential decay with rate λ. This is still an example of MCAR and can be thought to simulate assets failing to trade because of random shortages of liquidity;

3. **Stressed**: we calculate index squared returns for each date t and rank them. Then, returns whose date is above a certain percentile $p$get deleted with probability h, while the rest still go missing with rate r. This simulates shortages of liquidity or circuit breaks that happen when markets experience high levels of volatility; since the missingness of a given $r_{t,k}$ is injected conditional on all the other values of the dataset, this is an example of MNAR.

For each of the three missingness patterns, we will vary the parameters r, λ, and h to simulate different levels of missingness. Furthermore, we will make it so that the ratio of missing values is the same for each level of missingness in each pattern, so that we can compare the performance of the different imputation methods in a sensible way. The exact values of the parameters are reported in Table 2.2. The column "Rate" reports the expected value of the missingness ratio for each pattern.

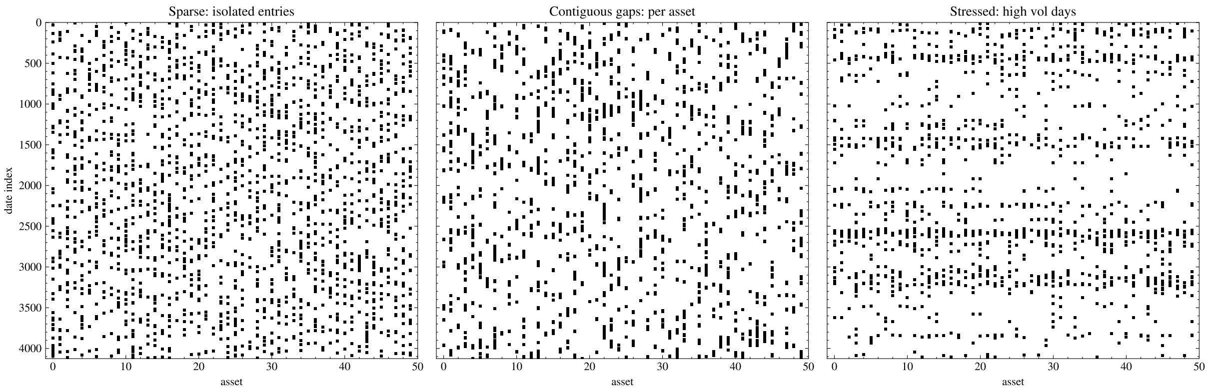

Figure 2.1: Missingness Examples

Sparses (r)	**Gaps** (n, λ)	**Stressed** (r, h, p)	**Rate**
0.001	(20, 0.1)	(0, 0.1, 0.99)	0.001
0.01	(200, 0.1)	(0.005, 0.5, 0.99)	0.01
0.05	(1000, 0.1)	(0.04, 0.2, 0.95)	0.05
0.1	(1000, 0.05)	(0.05, 0.5, 0.9)	0.1
0.3	(2000, 0.03)	(0.2, 1, 0.9)	0.3

Figure 2.2: Missingness parameters

2.2 Models

2.2.1 Deletion

Deletes rows where at least one missing value is found

2.2.2 Replacing with mean

Consists in setting each missing return to its expected value. This is almost equivalent to setting the missing price to be equal to the last observed price, since daily returns are very close to zero.

2.2.3 EM

With student-t

Dempster et al. (1977)

2.2.4 KNN

2.2.5 BRITS

Cao et al. (2018)

2.2.6 CSDI

2.3 Evaluation Methods

MSE

MAEa

CovErr =

$$\frac{\|\widehat{\Sigma}_{\text{imp}} - \Sigma_{\text{true}}\|}{\|\Sigma_{\text{true}}\|}$$

VaR

Minimum Variance Portfolio

2.4 Geometrical Interpretation

In this section we want to develop some geometrical intuition behind the evaluation methods criteria and the imputation models, as well as why a missing data problem requires different methods from an usual inference problem.

Take a multivariate time series of daily log returns:

$$R = \begin{pmatrix} r_1^\top \\ r_2^\top \\ \vdots \\ r_T^\top \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1N} \\ r_{21} & r_{22} & \cdots & r_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ r_{T1} & r_{T2} & \cdots & r_{TN} \end{pmatrix} \in \mathbb{R}^{T\times N}$$

The (t)-th observation is:

$$r_t = (r_{t1}, \ldots, r_{tN})^\top \in \mathbb{R}^N$$

The (i)-th variable asset realization is:

$$R_i = (r_{1i}, \ldots, x_{Ti})^\top \in \mathbb{R}^T$$

Let us first consider each asset separately. We interpret each asset return series as a finite-sample realization of an underlying square-integrable random variable in a Hilbert space:

$$R_i \in L^2(\Omega, \mathcal{F}, \mathbb{P}) = \left\{X : \Omega \to \mathbb{R} : \mathbb{E}[X^2] < \infty\right\}$$

Square integrability, which is a reasonable assumption [1], enables us to define an useful inner product on L^2, namely:

$$\langle X, Y\rangle =: \mathbb{E}[XY]$$

This is remarkably similar to the definition of covariance, and is equal when the mean of X and Y is zero; at the daily level this is almost the case. We can use this approximation to see that the covariance and correlation of two assets are a function of their angle in L^2:

$$\langle X, Y\rangle \simeq \mathrm{Cov}(X, Y)$$

$$\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)\,\mathrm{Var}(Y)} \simeq \frac{\langle X, Y\rangle}{\|X\|\|Y\|} = \cos(\theta)$$

The crucial part of this thesis is making sure that this linear geometry is preserved when imputing the missing data. The covariance matrix is of course the main object of

[1] Empirical work as Plerou et al. (1999) shows that the power law exponent of individual stocks is around 3, meaning variance exists but skew and kurtosis may not.

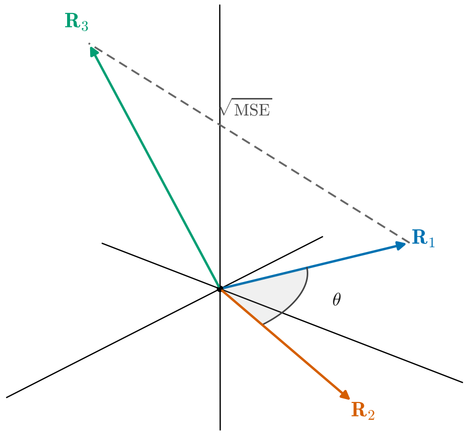

Figure 2.3: Returns as vectors in L^2

discussion, especially since it is used so widely across finance. It is immediate that MSE is the squared distance of two assets. The other metrics like VaR have no easy interpretation in this space.

Figure 2.3 is an attempt to represent this abstract space. Now let us turn to the realized space $\mathbb{R}^T$: the definitions above still apply, bar the due discretizations, but now we can represent the missing data as a projection of the full data. Write:

$$R_i = P_{\mathcal{O}_i} R_i + P_{\mathcal{M}_i} R_i = R_{O_i} + R_{M_i}$$

Where $P_{\mathcal{O}_i}$ is the operator that deletes the missing data and $P_{\mathcal{M}_i}$ is the one that keeps it and deletes the observed one. Their ranges are clearly orthogonal under the usual inner product. The goal of our models is to approximate R_{M_i} with:

$$\hat{R}_{M_i} = h_\theta(R_{O_i}, R_{O_{k \neq i}})$$

If we knew the values of R_{M_i} we could run an approximation in the least squares sense. Given h_θ, find:

$$\min_\theta \|R_{M_i} - \hat{R}_{M_i}\|$$

This is essentially equivalent to trying to approximate the conditional expectation, or in geometric terms projecting R_{M_i} onto the space of functions $\sigma(R_{O_i}$ and $R_{O_{k \neq i}})$, a subspace of L^2.

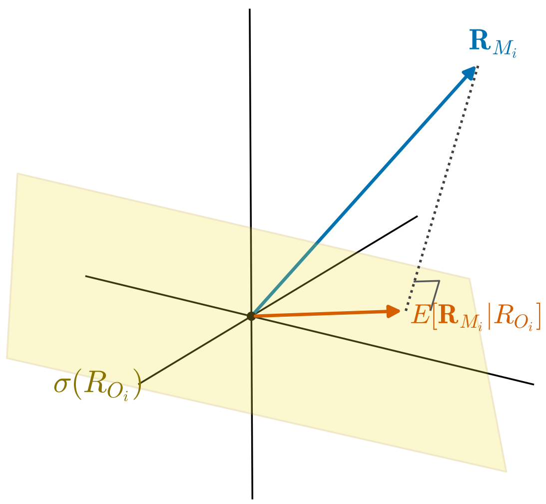

Figure 2.4: Conditional expectation as a projection

Chapter 3

Imputation Methods

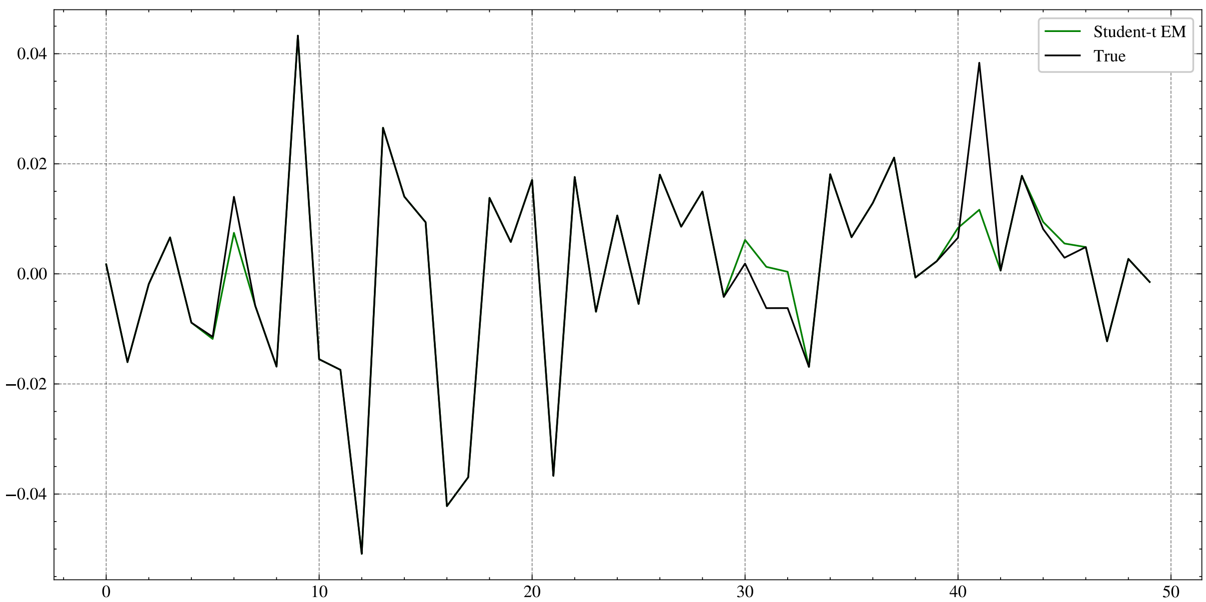

Figure 3.1: Examples of Imputed Data

Chapter 4

Results and Discussion

Chapter 5

Conclusion

5.1 Summary

5.2 Limitations

5.3 Future Work

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